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tuition

Angles & 2D Shapes

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: Two angles on a straight line are 48° and x°. What is x?
Method & answer: x = 180 − 48 = 132°.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What type of angle is 25°?

2. What type of angle is 150°?

3. Which type of angle is exactly 90°? A) acute B) obtuse C) right angle D) reflex

4. What type of angle is 250°?

5. What do angles on a straight line always add up to?

6. What do angles around a point always add up to?

7. How many sides does a heptagon have?

8. How many sides does a nonagon have?

9. Select ALL of these that are the sum of the interior angles of ANY quadrilateral: A) 360° B) 180° C) the same for every quadrilateral D) 90° E) different for a square vs a kite

10. How many sides does a decagon have?

11. What type of triangle has no equal sides?

12. What type of triangle has one angle greater than 90°?

13. How many lines of symmetry does a regular hexagon have?

14. What is the name for an angle greater than 180° but less than 360°?

15. How many sides does a triangle have?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Two angles on a straight line are 84° and x°. What is x?

2. Three angles around a point are 120°, 100° and x°. What is x?

3. A triangle has angles of 45° and 75°. What is the third angle? A) 50° B) 60° C) 70° D) 80°

6. In a triangle, two angles are equal at 48° each. What is the third angle?

7. Two straight lines cross, forming vertically opposite angles. If one angle is 63°, what is the angle vertically opposite it?

8. Select ALL of these that are TRUE about an isosceles triangle: A) it has exactly two equal sides B) it has exactly two equal angles C) all its angles are equal D) its base angles are equal E) it can also be right-

11. A quadrilateral has angles of 90°, 60°, and two other equal angles. What is the size of each of the other two?

12. Two angles on a straight line are in the ratio 3:7. What are the two angles?

13. The exterior angle of a triangle equals the sum of the two opposite interior angles. If those are 35° and 50°, what is the exterior angle?

angled

4. A quadrilateral has angles of 95°, 85° and 100°. What is the fourth angle?

5. Two equal angles lie around a point, and there are only two of them. What is the size of each?

9. An isosceles triangle has a base angle of 55°. What are the other two angles?

10. A right angle is split into two angles, one of which is 28°. What is the other angle?

14. A triangle has one angle of 90° and another of 25°. What is the third angle?

15. The interior angles of a regular octagon sum to 1080°. What is each interior angle?

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Angles a and b lie on a straight line, and a = 3b. Find b.

2. A regular polygon has an exterior angle of 24°. How many sides does it have? A) 12 B) 15 C) 18 D) 20

3. In a triangle, one angle is twice the second, and the third is 20° less than the second. Find the smallest angle.

4. Select ALL of these that are TRUE about co-interior (allied) angles on parallel lines cut by a transversal: A) they add to 180° B) they add to 360° C) they lie on the same side of the transversal D) they are always equal E) they only occur between parallel lines

5. A quadrilateral's angles are all equal. What is the size of each angle, and which shape has 4 equal angles?

6. Two parallel lines are cut by a transversal. Co-interior angles are (2x)° and (x+30)°. Find x.

7. A pizza is cut into equal slices, each with a 40° angle at the centre. How many slices are there in total?

8. A roof (viewed end-on) is an isosceles triangle with an apex angle of 100°. What angle does each sloping side make with the horizontal?

9. A door is opened so it makes an angle of 65° with the wall it is attached to. Through how many more degrees must it open to be at a right angle to the wall?

10. A pair of scissors opens to an angle of 38° between the blades. The angle is then increased by 15° three separate times. What is the final angle?

11. A regular polygon-shaped table has interior angles of 140° each. How many sides does the table have?

12. A triangular garden bed has one angle of 90° and another of 35°. What is the third angle, and what type of triangle is this (based on its angles)?

13. Two straight roads cross at a junction, forming four angles. One angle measures 115°. What are the sizes of the other three angles?

14. A kite-shaped flag has two equal angles of 75° each and one other angle of 130°. What is the size of the fourth angle?

15. A regular polygon has an interior angle of 144°. A second regular polygon has an interior angle exactly 36° smaller. How many sides does the second polygon have?